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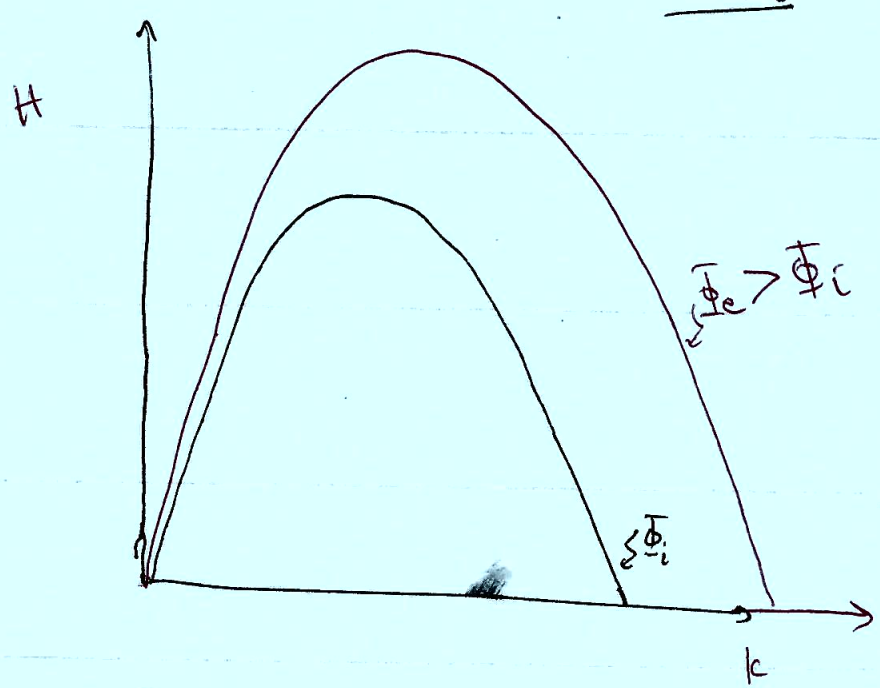
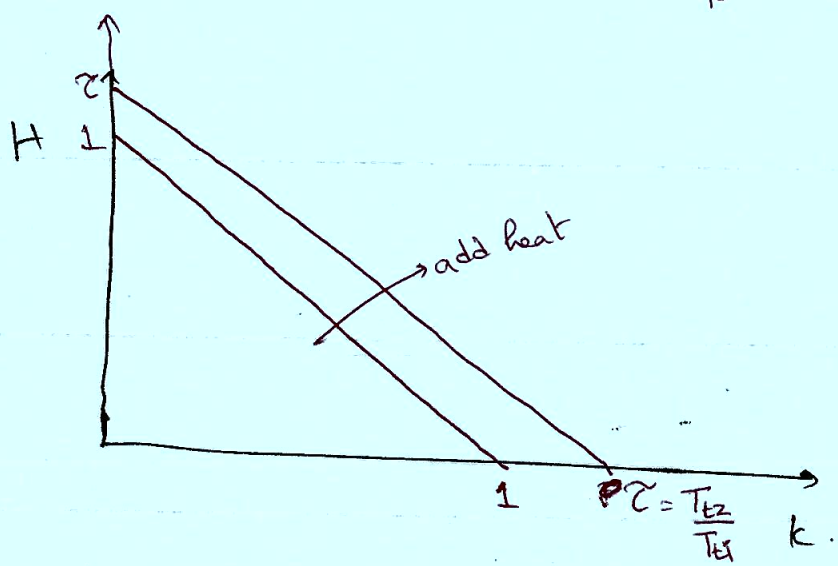
Lecture 8





$$H = \frac{C_p T}{C_p T_{ti}}$$

$$k = \frac{u^2}{2 C_p T_{ti}}$$

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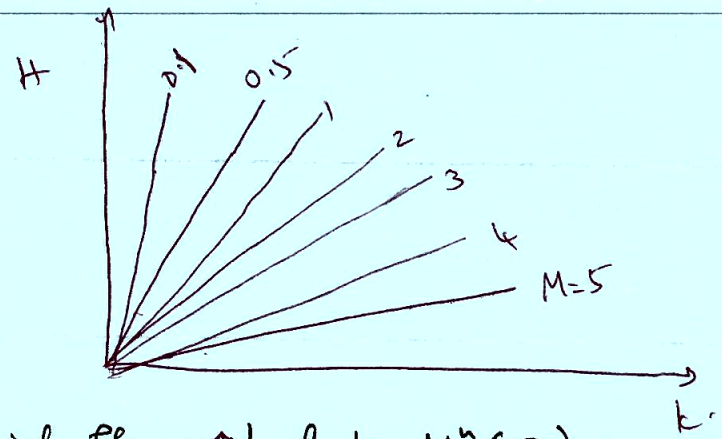
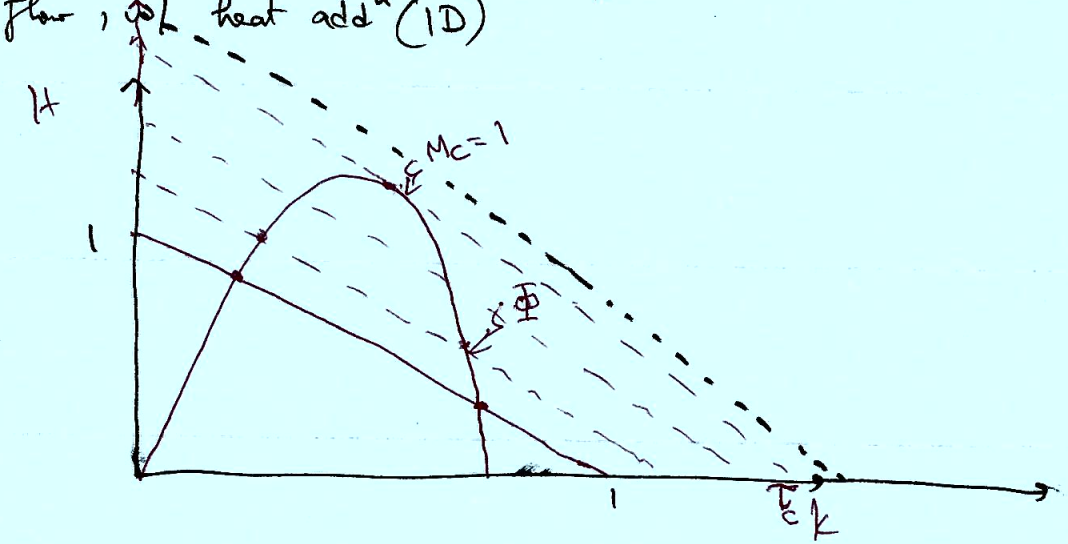


Consider Rayleigh flow, w/ heat add^n (1D)



Define

$$\tau_e = \frac{T_{te}}{T_{ti}} \Rightarrow \Phi = \left[\frac{\tau_e(\gamma-1)M^2}{1+\left(\frac{\gamma-1}{2}\right)M^2}\right]^{1/2}\left[1+\frac{1}{M^2\gamma}\right]$$

Can define $\tau_c \equiv$ critical total temp ratio for which we achieve a choked condition $M_c = 1$

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Can show

$$\tau_c = \frac{\Phi^2}{2\left(1-\frac{1}{\gamma^2}\right)}$$

If we add more heat, we must change inlet conditions if we cross the choking pt.

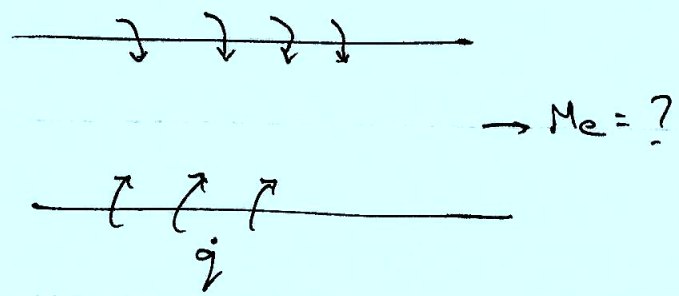
Ex: Rayleigh flow with $\tau = 2.5$ (Note $\tau = \frac{T_{te}}{T_{ti}}$)

$\gamma = 1.4$

$C_p = 1000$ J/kgK

$T = 1000$ K

$V = 100$ m/s



$\rightarrow M_e = ?$

Φ = const (b/c 1D, const area, heat add^n)

$$\Phi = \frac{1}{\sqrt{2k}}\left(2k + \frac{\gamma-1}{\gamma}H\right)$$

$$= \frac{1}{2}\left(\frac{2C_pT_{ti}}{u^2}\right)^{1/2}\left(\frac{u^2}{C_pT_{ti}} + \frac{\gamma-1}{\gamma}\frac{C_pT}{C_pT_{ti}}\right)$$

$T_{ti} = ?$ $u_i = 100$ m/s.

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$$a_i = \sqrt{\gamma R T_i} = \sqrt{1.4 \times 287 \times 1000}$$

$$M_i = \frac{u_i}{a_i} = \frac{100}{\sqrt{1.4 \times 287 \times 1000}} = 0.158$$

$$T_{ti} = 1005\,K.$$

$$T_i = 1000\,K$$

$$H_i = \frac{C_p T_i}{C_p T_{ti}} = 0.995$$

$$k_i = \frac{u^2}{2 C_p T_{ti}} = 0.005$$

$$\Phi = \frac{1}{\sqrt{2k}} \cdot \left(2k + \frac{\gamma-1}{\gamma} H\right) = \frac{1}{\sqrt{2 \times 0.005}} \left(2 \times 0.005 + \frac{1.4-1}{1.4} \times 0.99\right)$$

To start with, before heat add^n

$$\Phi = 2.94$$

$$\Phi_c = 0.99 \quad \text{for } \gamma = 1.4$$

$$\Rightarrow \Phi > \Phi_c \Rightarrow 2 \text{ sol}^n\text{s.}$$

$$K_e = \left\{ \frac{\gamma \Phi}{\sqrt{2}(\gamma+1)} \pm \left(\frac{1}{2}\left(\frac{\gamma \Phi}{\gamma+1}\right)^2 - \tau \left(\frac{\gamma-1}{\gamma+1}\right) \right)^{1/2} \right\}^2$$

($\Phi \to 2.94$, $\gamma \to 1.4$, $\tau \to 2.5$)

$$\boxed{K_e = \frac{u_e^2}{2 C_p T_{ti}}}$$

$$V_e = 260 \text{ m/s}$$

Alternatively after heat add^n.

$$\tau_c = \frac{\Phi^2}{2\left(1 - \frac{1}{\gamma^2}\right)} = \frac{2.94^2}{2\left(1 - \frac{1}{1.4^2}\right)}$$

$$\tau_c = 8.8 = \frac{T_{te,c}}{T_{ti}} \Rightarrow T_{te,c} \simeq 9000\,K$$

$$\tau_{given} = 2.5 < \tau_c$$

$$\Rightarrow 2 \text{ sol}^n\text{s.}$$

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$k_e = 0.033$

$\Rightarrow H_e = \tau_e - k_e$

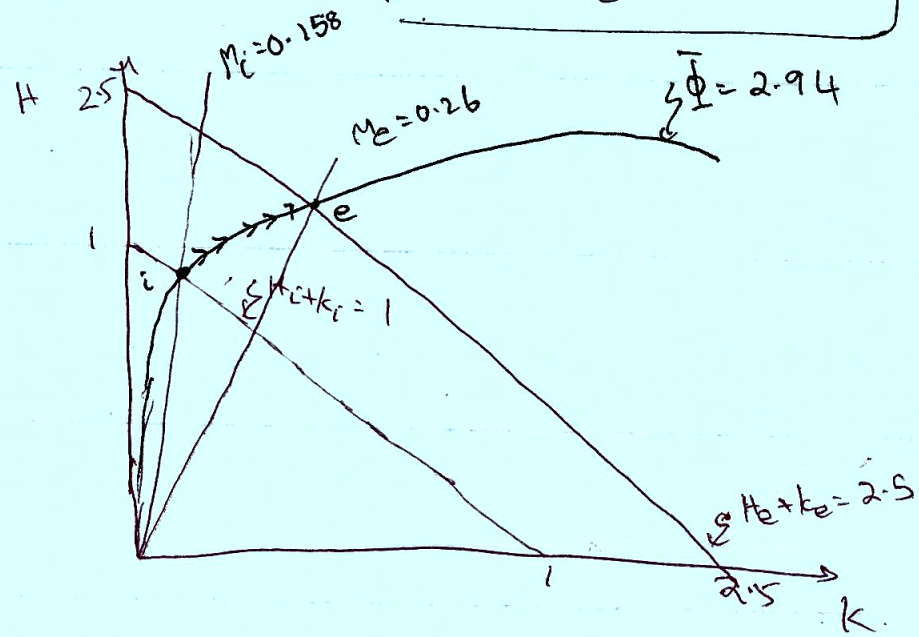
$= 2.5 - 0.033$

$\boxed{H_e = 2.47}$

$\Rightarrow H_e = \dfrac{C_p T_e}{C_p T_{bi}} \Rightarrow T_e = 2.47 \times 1005 = 2478\,K.$

$a_e = \sqrt{\gamma R T_e} = 998\,m/s$

$\boxed{M_e = \dfrac{u_e}{a_e} = 0.26}$



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In Rayleigh flow, heat is added in a constant area duct (frictionless)

In Brayton cycle, heat is added isobarically ($p = \phi$). What does isobaric heat addition look like in H-k space?

Consider frictionless flow thru a <u>variable</u> area duct w/ heat add^n.

1st law in differential form

$$\dot{m}(dh + VdV) = \delta\dot{W}_o + \delta\dot{Q}_o + \underbrace{d\dot{m}_b\left(h_b - h + \frac{V_b^2 - V^2}{2}\right)}_{\text{add}^n \text{ thru walls}}$$

Let $\delta\dot{W}_o = 0$.

Mom. eq^n

$$dp + \rho V dV = \frac{1}{A} dF_{b,x} + \left(\frac{V_b - V}{A}\right) d\dot{m}_b$$

Subscript 'b' refers to mass flux thru walls & $dF_{b,x}$ are shear forces & drag of any ~~protrs~~ proturberances.

If the walls are frictionless, & no proturberances

$$dF_{b,x} = 0$$

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If no mass add^n thru walls,

$$d\dot{m}_b = 0$$

So cons. of mom. reduces to

$$dp + \rho V dV = 0$$

$$\cancel{dp = \rho V dV} \quad \cancel{(\text{Euler eq}^n)}$$

$$dp = -\rho V dV \quad (\text{Euler eq}^n)$$

If we add heat isobarically, $dp = 0$

$$\rho V dV = 0$$

$$V dV = 0$$

$$d\left(\frac{V^2}{2}\right) = 0$$

$$\boxed{dk = 0} \Rightarrow k = \text{const.}$$

Two ways to add heat (ideally)

a) Const area (Rayleigh flow)

$$\Phi = \phi$$

k can increase (if subsonic) or decrease (if supersonic)

H must increase.

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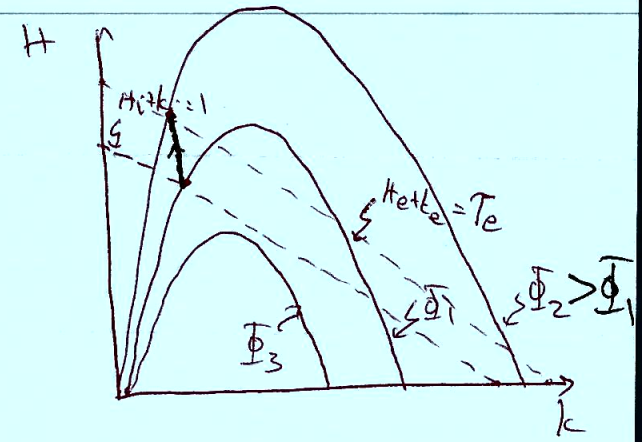
(b) Isobaric heat addⁿ (Brayton cycle)

$$k = \text{¢} \; ; \; H \uparrow$$

$$\bar{\Phi} : \uparrow$$



For adiabatic flow w/ area change

$$\bar{\Phi} = f(\gamma, M)$$

For non-adiabatic flow ($T \neq 1$) w/ area change

$$\bar{\Phi} = \left\{ \frac{T(\gamma-1)M^2}{1+\frac{\gamma-1}{2}M^2} \right\}^{1/2} \left(1 + \frac{1}{\gamma M^2}\right)$$

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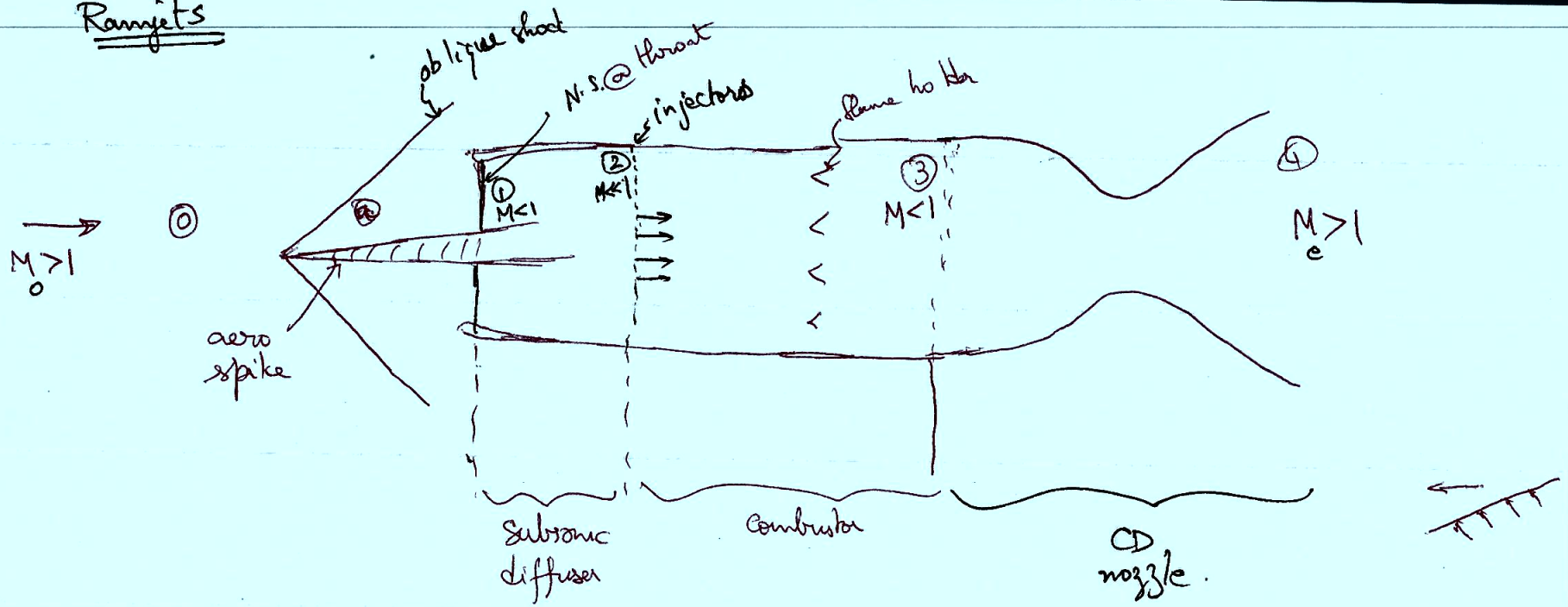
Ramjets



Recall for ideal ramjet $M_e = M_0$ & $p_t = \phi$

$$V_e = \sqrt{\frac{T_e}{T_0}}\, u_0$$

$0 \to 1$: flow goes thru oblique & normal shock.

Across normal shock $\Phi = \phi$

Across oblique shock $\Phi \downarrow$ because the flow now goes thru an area change!

$1 \to 2$: Subsonic deceleration $\Phi \uparrow$

$2 \to 3$ Isobaric heat add $(p = \phi,\ \Phi \uparrow)$

$3 \to 4$ accel to M_e ($\Phi \downarrow$ in converging ; $\Phi \uparrow$ in diverging)

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